

42) Show that the logical equivalences in Table 6, except for the double negation law, come in pairs, where each pair contains compound propositions that are duals of each other.

Soln:

Identity Laws

$$p \wedge T \equiv p$$

$$p \vee F \equiv p$$

$$\text{Dual: } p \vee F \equiv p$$

$$\downarrow \text{Dual}$$

$$p \wedge T \equiv p$$

Domination Laws

$$p \vee T \equiv T$$

$$p \wedge F \equiv F$$

$$\text{Dual: } p \wedge F \equiv F$$

$$\text{Dual: } p \vee T \equiv T$$

Idempotent Laws

$$p \vee p \equiv p$$

$$p \wedge p \equiv p$$

$$\text{Dual: } p \wedge p \equiv p$$

$$\text{Dual: } p \vee p \equiv p$$

Commutative Laws

$$p \vee q \equiv q \vee p$$

$$p \wedge q \equiv q \wedge p$$

$$\text{Dual: } p \wedge q \equiv q \wedge p$$

$$\text{Dual: } p \vee q \equiv q \vee p$$

Associative Laws

$$(p \vee q) \vee r \equiv p \vee (q \vee r)$$

$$(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$$

$$\text{Dual: } (p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$$

$$\text{Dual: } (p \vee q) \vee r \equiv p \vee (q \vee r)$$

Distributive Laws

$$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$$

$$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$$

$$\text{Dual: } p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$$

$$\text{Dual: } p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$$

De Morgan's Laws

$$\neg(p \wedge q) \equiv \neg p \vee \neg q$$

$$\neg(p \vee q) \equiv \neg p \wedge \neg q$$

$$\text{Dual: } \neg(p \vee q) \equiv \neg p \wedge \neg q$$

$$\text{Dual: } \neg(p \wedge q) \equiv \neg p \vee \neg q$$

Absorption Laws

$$p \vee (p \wedge q) \equiv p$$

$$p \wedge (p \vee q) \equiv p$$